

COM3031 Probabilistic Machine Learning: Week 6

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In this week

Lectures:

- Measuring Uncertainty
- Information Flow Between Variables
- Comparing Probability Distributions

Workshop:

- KL divergence and ELBO

Part I: Measuring Uncertainty

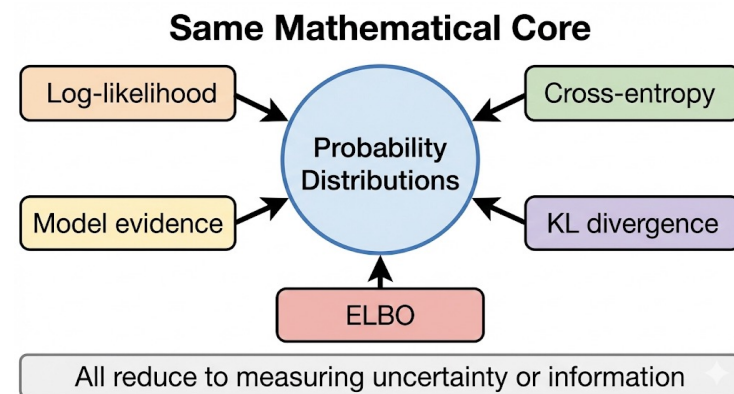
A Hidden Quantity in Probabilistic Machine Learning

What We Have Been Doing:

- Maximised log-likelihood
- Used cross-entropy loss for classification
- Minimized KL divergence in Variational Inference
- Maximised the ELBO
- Compared models using log model evidence
- Sampled from posterior distributions

Notice something:

- Logarithms appear everywhere
- We compare probability distributions
- We measure uncertainty repeatedly



Question: What is the fundamental quantity that connects all of these ideas?

Log-Likelihood, loss and KL divergence

Log-likelihood:

$$\log p(\mathcal{D}|\theta) = \sum_{i=1}^n \log p(y_i|\theta)$$

Interpretation:

- Each term measures how “surprised” the model is
- We maximise log-likelihood
- Equivalent to minimising negative log-likelihood

Logistic Regression:

$$-\sum_i y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)$$

- This is **cross-entropy loss**.
- We didn’t introduce it randomly. It comes from a deeper information measure.

Variational Inference:

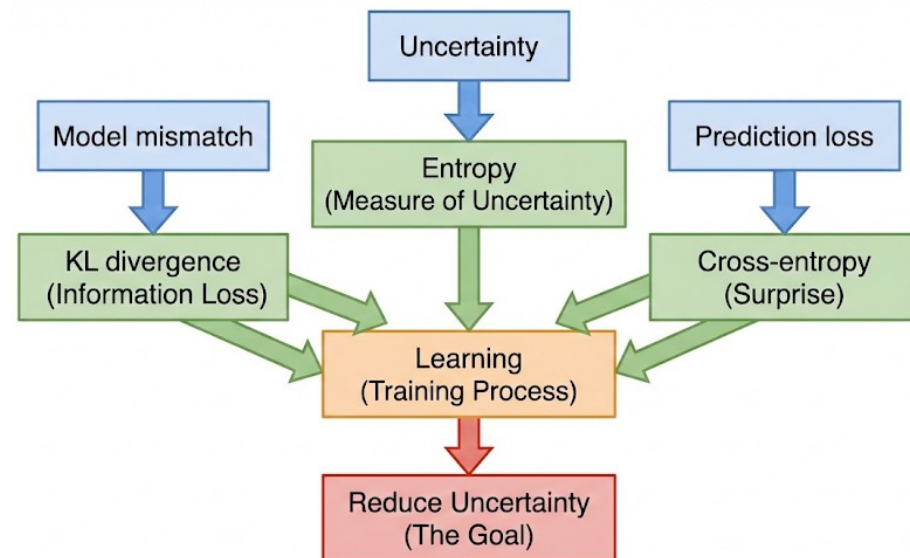
$$\text{KL}(q(\theta) || p(\theta | \mathcal{D}))$$

- In model comparison: $\text{KL}(p|q)$
- We’ve repeatedly compared probability distributions.
- These are not random choices. They are all measures of **information**.

Information Theory: The Language of Uncertainty

Everything Is Measuring Information

- Cross-entropy = expected surprise under data
- KL divergence = extra surprise due to mismatch
- ELBO = balancing data fit and uncertainty
- Model evidence = total information in data



What Is Information?

Information = Surprise

When we observe an outcome x ,

- If it was very likely $\rightarrow$ little new information
- If it was unlikely $\rightarrow$ large information
- If it was certain ($p(x)=1$) $\rightarrow$ no information

A Quantitative Measure of Surprise

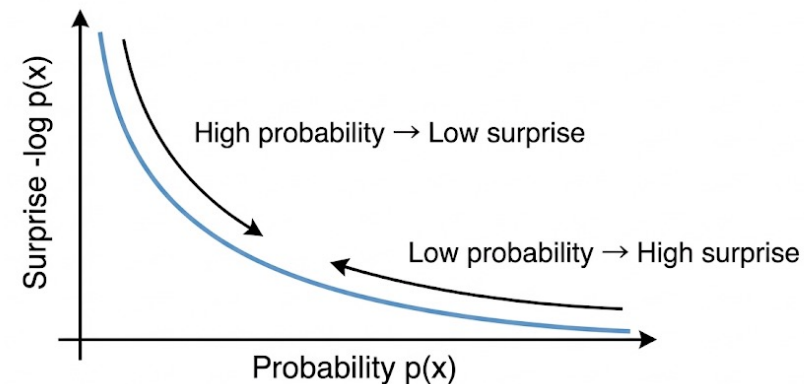
We define the information content of outcome x as:

$$I(x) = -\log_2 p(x)$$

- $\log_2 \rightarrow$ information measured in bits
- $\ln \rightarrow$ nats
- Additivity: for independent events x and y :
 $I(x, y) = I(x) + I(y)$
- Negative sign ensures information measure is positive

Information measures *how unexpected* an outcome is.

Surprise Curve



Learning = Minimising total surprise.

From Information to Entropy

Single outcome information:

$$I(x) = -\log_2 p(x)$$

- Information measures how surprising one outcome is.

If $X \sim p(x)$, the **average information** is:

$$H(X) = \mathbb{E}_{\mathbb{P}}[I(X)]$$

Substitute $I(X)$:

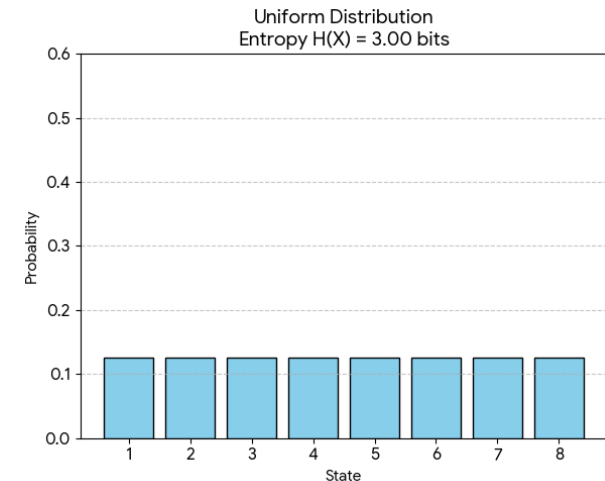
$$H(X) = \mathbb{E}_{\mathbb{P}}[-\log_2 p(X)] = -\sum_x p(x) \log_2 p(x)$$

- This is called **Entropy** (discrete case).
- **Entropy = average surprise**
- Measures overall uncertainty of a distribution
 - Low entropy $\rightarrow$ low uncertainty and High entropy $\rightarrow$ high uncertainty
- Not about one outcome– about the whole distribution

Entropy for Discrete Random Variables

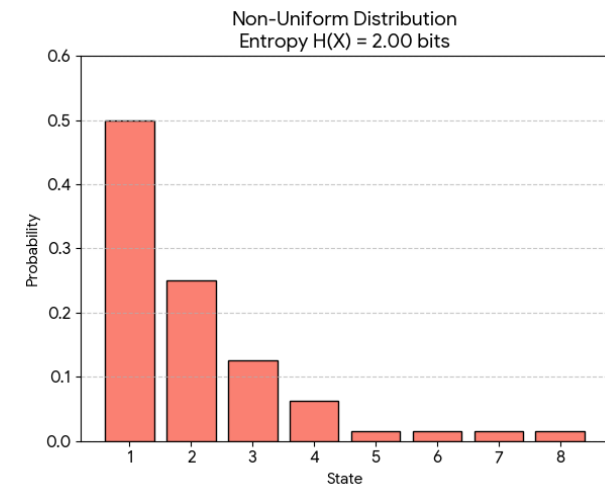
Uniform case:

- If X has K equally likely outcomes: $p(x) = \frac{1}{K}$.
- For example, 8 equally likely states: $H(X) = -8 \times \left(\frac{1}{8}\right) \log_2 \left(\frac{1}{8}\right) = 3 \text{ bits}$
- Entropy is maximized when the distribution is uniform.



Non-uniform case:

- If X has 8 states with probabilities: $(1/2, 1/4, 1/8, 1/16, 1/64, 1/64, 1/64, 1/64)$
- $H(X) = 2 \text{ bits}$
- Non-uniform distribution has smaller entropy than uniform distribution.



Entropy and Prefix-Free Codes

What does “average surprise” mean in practice?

- To encode outcomes of a discrete random variable X using the minimum average number of bits.

Prefix-free codes (so decoding works)

- A code is prefix-free if no codeword is the start of another.
- This guarantees unique and instant decoding.
- Example (prefix-free):
 $\{0, 10, 110, 1110, 111100, 111101, 111110, 111111\}$
- Prefix-free means we can decode greedily, left-to-right, without ambiguity.

Probability $\leftrightarrow$ code length (optimal rule)

In an efficient prefix-free code:

$$\ell(x) \approx -\log_2 p(x)$$

- High probability $\rightarrow$ short code
- Low probability $\rightarrow$ long code

Average Code Length:

$\mathbb{E}[\ell(X)] = \sum_x p(x)\ell(x)$ (average number of bits used)

- Shannon’s Noiseless Coding Theorem:

$\mathbb{E}[\ell(X)] \geq H(X)$ and with codes are optimal: $\mathbb{E}[\ell(X)] \approx H(X)$

Entropy and Average Code Length

Example

- Suppose X has 8 outcomes with probabilities:
 $(1/2, 1/4, 1/8, 1/16, 1/64, 1/64, 1/64, 1/64)$
- We assign shorter codes to more probable outcomes.
- Average code length:
 $(1/2) \times 1 + (1/4) \times 2 + (1/8) \times 3 + (1/16) \times 4$
 $+ 4 \times (1/64) \times 6 = 2 \text{ bits}$
 - Equal to the entropy $H(X)$

Key insights:

- Entropy is not abstract.
- It is a **fundamental limit of compression**.
- In **machine learning**:
 - Cross-entropy = coding cost under wrong model
 - Log-likelihood = negative coding length
 - KL divergence = extra coding cost

Everything we do is about **efficient encoding of uncertainty**.

Maximum Entropy Principle

Key question: If entropy measures uncertainty, which distribution should we choose when we only know *partial* information?

What we know (constraints):

We often know only a few facts about a random variable X , e.g.,

- Support: $x \in \mathcal{X}$
- Mean: $\mathbb{E}[X] = \mu$
- Variance: $\text{Var}(X) = \sigma^2$
- ...but we *don't* know the full distribution $p(x)$.

Core principle (least-assumptive choice):

- Among all distributions $p(x)$ that satisfy the constraints, choose the one with **maximum entropy**:

$$p^* = \arg \max_{p \in \mathcal{C}} H(p) \quad \text{where} \quad H(p) = - \sum_{x \in \mathcal{X}} p(x) \log p(x)$$

- $\mathcal{C}$ is the set of distributions that satisfy what we know

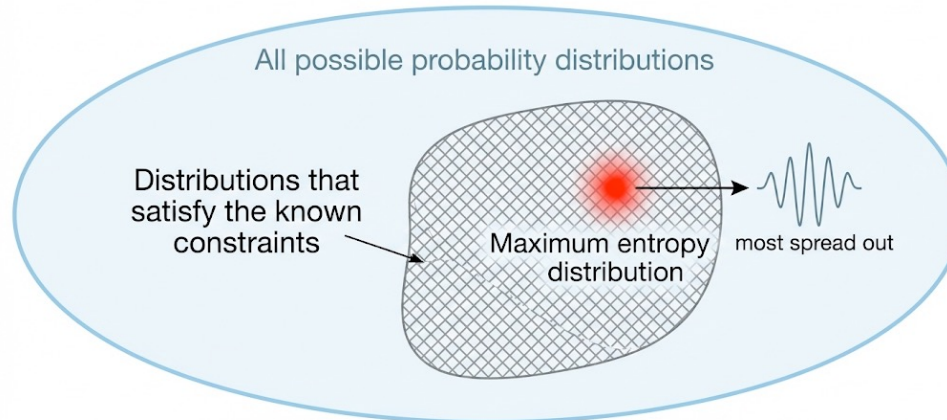
Maximum Entropy Principle

Why maximise entropy?

- Max entropy = most spread out / most uncertain distribution allowed by the constraints
- Adds no extra structure beyond the facts we're given
- Acts as a principled default (“least biased” model consistent with the evidence)

Key takeaway: Max entropy chooses the distribution that matches the constraints while making the fewest additional assumptions.

Among all feasible distributions, choose the one that is most spread out.



Maximum Entropy: How Do We Find It?



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Goal: Find the distribution $p(x)$ that maximizes entropy and satisfies the probability constraint $\sum_i p(x_i) = 1$.

Step 1: Set up the optimisation problem

- Maximise $H(p) = -\sum_i p(x_i) \ln p(x_i)$ subject to $\sum_i p(x_i) = 1$.

Step 2: Use a Lagrange multiplier

- Introduce λ to enforce the constraint:

$$\mathcal{L}(p, \lambda) = -\sum_i p(x_i) \ln p(x_i) + \lambda(\sum_i p(x_i) - 1)$$

What happens next? (to do on board)

- Take derivative w.r.t. $p(x_i)$
- Set equal to zero
- Solve for $p(x_i)$

Maximum Entropy: How Do We Find It?



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Result (Discrete Case)

- If X has K possible outcomes and we only require $\sum_i p(x_i) = 1$
- then the entropy is maximised when

$$p(x_i) = \frac{1}{K} \quad \text{for all } i$$

- That is, the **uniform distribution**.

Maximum Entropy Value

$$H_{\max} = - \sum_{i=1}^K \frac{1}{K} \ln \frac{1}{K} = \ln K$$

So entropy increases with the number of possible states.

Key Takeaway

- With no additional constraints, the least biased choice is uniform.
- Maximum entropy = maximum uncertainty.
- Add constraints $\rightarrow$ the max-entropy solution changes.

Maximum Entropy with Constraints

What changes when we know something?

- Suppose we also know:

$$\mathbb{E}_{\mathbb{P}}[f(X)] = \mu$$

- That is, the distribution must satisfy a moment constraint.

New optimisation problem

- Maximise entropy:

$$H(p) = -\sum_x p(x) \ln p(x) \text{ subject to } \sum_x p(x) = 1 \text{ and } \sum_x p(x)f(x) = \mu$$

- After applying Lagrange multipliers, the maximum-entropy distribution becomes:

$$p(x) = \frac{1}{Z(\lambda)} \exp(\lambda f(x))$$

- This is an **exponential family distribution**.

Why this matters for ML

- Logistic regression, Softmax classifiers, Exponential family models Energy-based models. All follow this form.

Entropy for Continuous Random Variables

Entropy for Discrete Variables (recap)

$$H(X) = - \sum_x p(x) \log p(x)$$

Measures average surprise.

What changes for continuous variables?

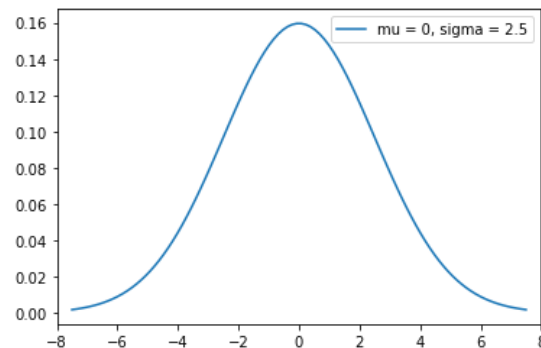
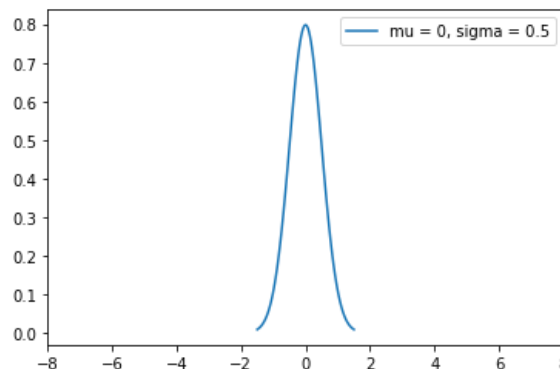
Let $X \sim p(x)$ be continuous, we define **differential entropy**:

$$h(X) = - \int p(x) \log p(x) dx$$

Key Differences from Discrete Entropy

- Can be **negative**
- Depends on the scale (not invariant to reparameterisation)
- Measures spread in a relative sense

Narrow Gaussian → low uncertainty → low entropy



Wide Gaussian → high uncertainty → high entropy

The Maximum Differential Entropy

Questions: Among all continuous distributions satisfying given constraints, which one maximizes entropy?

Suppose we know:

$$\mathbb{E}[X] = \int x p(x) dx = \mu. \quad \text{Var}(X) = \int (x - \mu)^2 p(x) dx = \sigma^2$$

- but we do NOT know the full distribution.

The Maximum Entropy Solution

By maximizing differential entropy under constraints on the mean and variance using Lagrange multipliers, the resulting distribution turns out to be **Gaussian**:

$$X \sim \mathcal{N}(\mu, \sigma^2), \quad p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

- It is the **most uncertain distribution** under those constraints.
- It adds no assumptions beyond the specified mean and variance.

Differential Entropy of A Gaussian

Step 1: Differential Entropy Definition

- For a continuous random variable:

$$h(X) = - \int p(x) \log p(x) dx = \mathbb{E}[-\log p(X)]$$

Step 2: Gaussian log-density

- If $X \sim \mathcal{N}(\mu, \sigma^2)$,

$$\log p(x) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(x - \mu)^2}{2\sigma^2};$$

So,

$$-\log p(X) = \frac{1}{2} \log(2\pi\sigma^2) + \frac{(X - \mu)^2}{2\sigma^2}$$

Step 3: Take expectation:

$$h(X) = \mathbb{E}[-\log p(X)] = \frac{1}{2} \log(2\pi\sigma^2) + \frac{1}{2\sigma^2} \mathbb{E}[(X - \mu)^2]$$

Differential Entropy of A Gaussian

Step 4: Use the variance identity:

$$\mathbb{E}[(X - \mu)^2] = \text{Var}(X) = \sigma^2$$

- Substitute:

$$h(X) = \frac{1}{2} \log(2\pi\sigma^2) + \frac{1}{2}$$

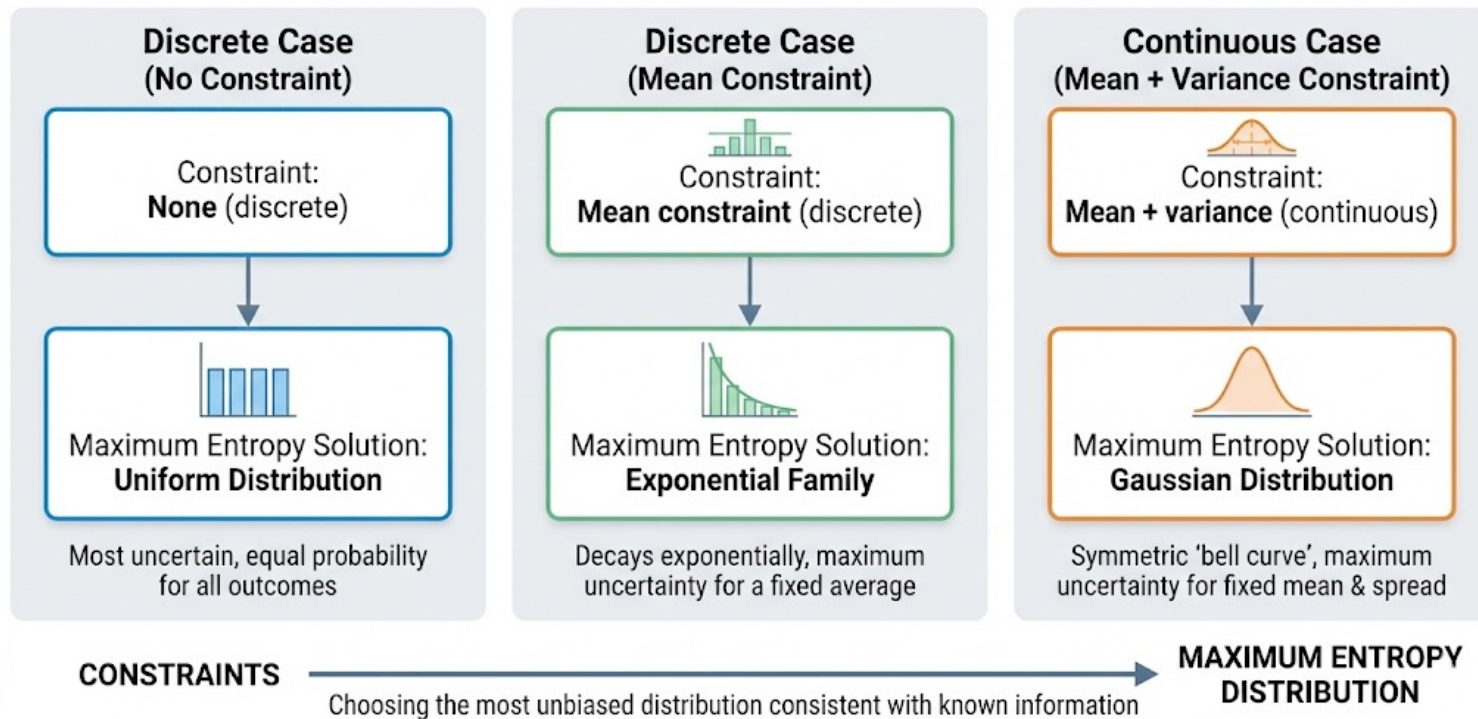
Final Result:

$$h(X) = \frac{1}{2} \log(2\pi e\sigma^2)$$

Interpretation

- Entropy increases as variance increases
- Wider Gaussian $\rightarrow$ more spread $\rightarrow$ larger uncertainty $\rightarrow$ larger entropy
- Differential entropy depends on scale

Maximum Entropy Principle: Constraints & Solutions



Part II: Information Flow Between Variables

Conditional Entropy

Motivation

- Entropy $H(X)$ measures uncertainty in a variable X . But often we know something else.
- How uncertain is X **after observing** Y ?

Information Perspective

If we observe a pair $(x, y) \sim p(x, y)$,

- The information needed to specify x given y is: $I(x | y) = -\log p(x | y)$

So the **average remaining uncertainty** in X after observing Y is:

$$H(X | Y) = \mathbb{E}_{p(x,y)}[-\log p(x | y)] \text{ or } H(X | Y) = \sum_{x,y} p(x, y) \cdot (-\log p(x | y))$$

Interpretation

- Measures **remaining uncertainty in** X once Y is known.
- If Y completely determines $X \rightarrow H(X | Y) = 0$
- If Y gives no information about $X \rightarrow H(X | Y) = H(X)$

Conditional Entropy

Key Identity (Chain Rule of Entropy)

$$H(X, Y) = H(Y) + H(X | Y)$$

- Total uncertainty in (X, Y) = uncertainty in Y + remaining uncertainty in X after observing Y

In Machine Learning:

- **Classification:** X = true label; Y = input features
 - $H(X | Y)$ measures how uncertain the label is **after seeing the features**
 - Small $H(X | Y) \rightarrow$ features strongly determine the label
 - Large $H(X | Y) \rightarrow$ features leave ambiguity
 - A perfect classifier would make $H(X | Y) \rightarrow 0$
- **Learning** = reducing conditional entropy $H(X|Y)$

Mutual Information

Motivation:

- Entropy $H(X)$: uncertainty in X
- Conditional entropy $H(X | Y)$ measures remaining uncertainty after observing Y .

Question: how much does observing Y reduce uncertainty about X ?

Mutual Information:

$$I(X; Y) = H(X) - H(X|Y)$$

- This means uncertainty removed about X after observing Y .

Bayesian Perspective

- In Bayesian inference: $X \rightarrow$ parameters θ and $Y \rightarrow$ data $\mathcal{D}$

$$I(\theta; \mathcal{D}) = H(\theta) - H(\theta|\mathcal{D})$$

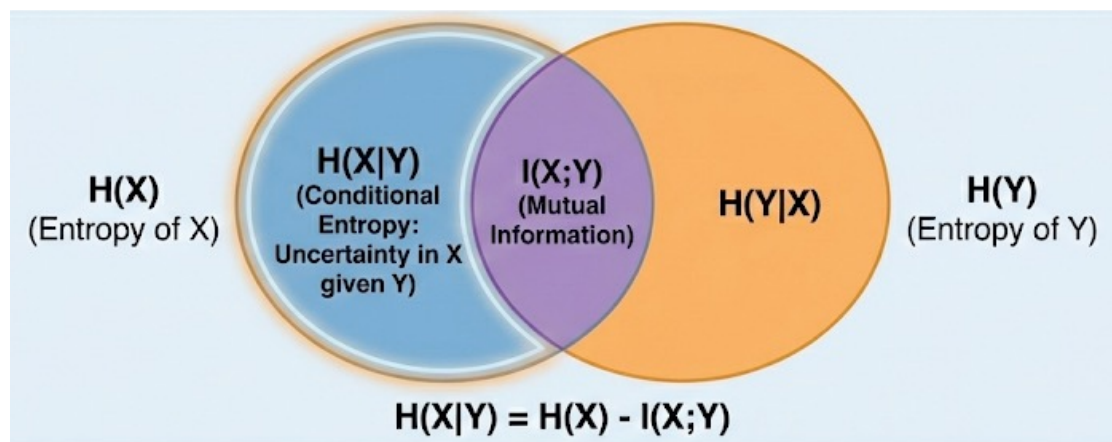
- How much information the data provides about the parameters.
- Large $I(\theta; \mathcal{D}) \rightarrow$ data strongly constrains the model
- Small $I(\theta; \mathcal{D}) \rightarrow$ data provides little information
- Bayesian learning = reducing uncertainty/entropy about parameters using data.

Mutual Information

Mutual Information: Equivalent Forms

$$I(X; Y) = H(X) - H(X|Y) = D_{KL}(p(x, y) | p(x)p(y))$$

- Mutual information measures deviation from independence.
- If X and Y are independent:
$$p(x, y) = p(x)p(y) \Rightarrow I(X; Y) = 0$$
- Larger $I(X; Y) \rightarrow$ stronger statistical dependence.



Part III: Comparing Probability Distributions

Relative Entropy

Motivation

- Entropy measures uncertainty of one distribution. But in machine learning, we often want to compare two distributions.

Question:

- How different is a model distribution $q(x)$ from the true distribution $p(x)$?

$$D_{\text{KL}}(p|q) = \sum_x p(x) \log \frac{p(x)}{q(x)}$$

Expand the log ratio:

$$D_{\text{KL}}(p|q) = \sum_x p(x) \log p(x) - \sum_x p(x) \log q(x)$$

- Rearrange: $D_{\text{KL}}(p|q) = -H(p) + H(p, q)$

Cross-Entropy:

$$H(p, q) = - \sum_x p(x) \log q(x)$$

- Expected surprise when data is generated from p , but encoded using q .

Relative Entropy

Relationship Between the Three Quantities

$$H(p, q) = H(p) + D_{\text{KL}}(p|q)$$

- Entropy $\rightarrow$ intrinsic uncertainty of p
- Cross-entropy $\rightarrow$ coding cost using q
- KL divergence $\rightarrow$ extra cost due to mismatch

Why This Matters in Machine Learning

- In classification: p = true label distribution, q = model prediction

$$\mathcal{L} = -\sum_i y_i \log \hat{y}_i \text{ (cross-entropy loss)}$$

If labels are one-hot:

- Minimising cross-entropy $\equiv$ Minimising KL divergence between true and predicted distributions
- Because: $H(p)$ is fixed w.r.t model parameters
- So: Minimise $H(p, q) \Leftrightarrow$ Minimise $D_{\text{KL}}(p|q)$

Summary

Information as Surprise

- Information content: $I(x) = -\log p(x)$
- Rare events $\rightarrow$ high surprise
- Likely events $\rightarrow$ low surprise

Entropy = Average Uncertainty

- Discrete entropy and Continuous (differential) entropy
- Measures unpredictability of a distribution

Maximum Entropy Principle

- No constraints $\rightarrow$ Uniform
- Fixed mean $\rightarrow$ Exponential family
- Fixed mean & variance $\rightarrow$ Gaussian

Relative Entropy (KL Divergence)

- $D_{KL}(p|q) = H(p, q) - H(p)$
- Measures mismatch between distributions. Appears in:
- Variational inference
- Model comparison
- Cross-entropy loss

Conditional Entropy & Mutual Information

- Conditional entropy:
Remaining uncertainty after observing Y
- Mutual information:
- $I(X; Y) = H(X) - H(X|Y)$
- Information gained about X from Y